Ayush Agrawal | ECE M237 (Dynamic Programming) | UCLA Spring 2026

RISK-AWARE LANE-CHANGE PLANNING

via Online CVaR-Augmented MPPI

Motivation: Why do we need Conditional Value at Risk (CVaR)?

- Highway lane-changes: Planner generates hundreds of routine outcomes for every catastrophic one.
- Standard **Model Predictive Path Integral** (MPPI) averages over all of them. The catastrophic 1-in-20 is invisible.
- The right mathematical object for that tail is **Conditional Value at Risk (CVaR)**.

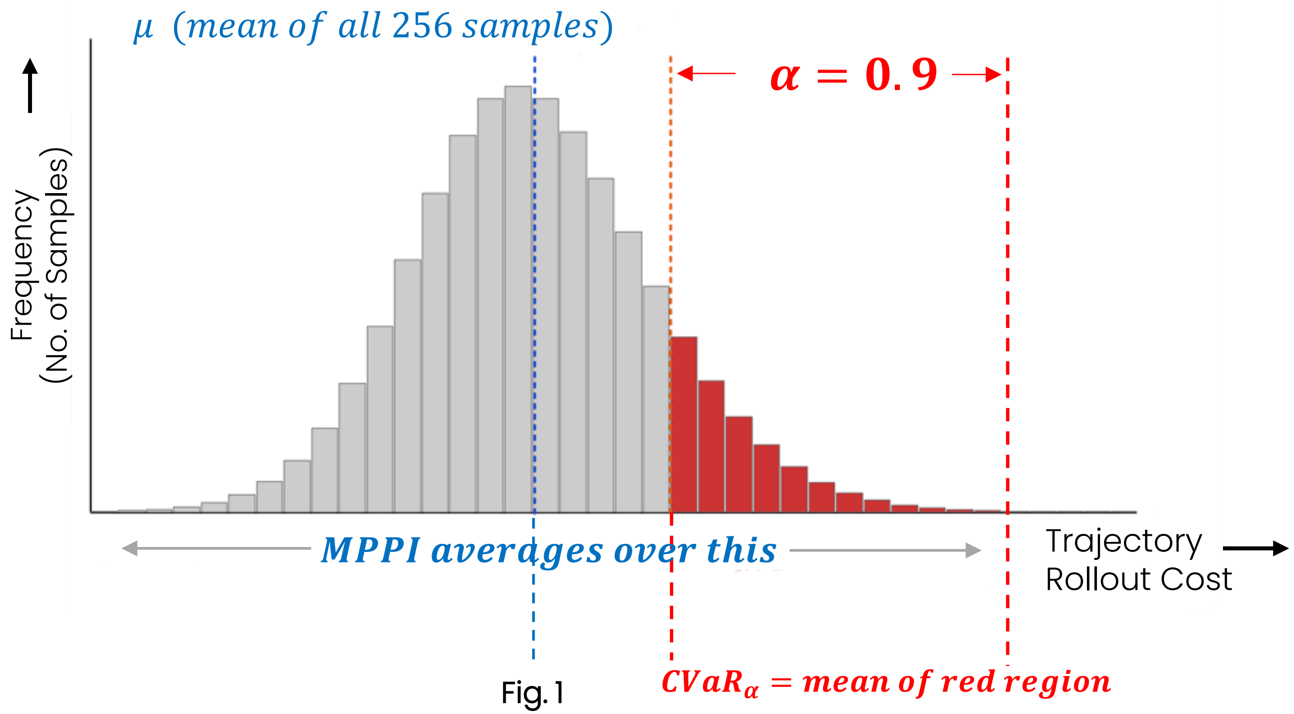


Fig. 1

CvaR makes the planner aware about risky situations resulting in a conservative decision making leading to more safety.

Problem Formulation

Variables and Expressions	Description
$x_k \in R^{n_x}$	the joint ego-traffic state
$u_k = v_k + \epsilon_k \in R^{n_u}$	the perturbed ego control with $\epsilon_k \sim \mathrm{N}(0, \Sigma_\epsilon)$.
w_k	is exogenous traffic noise
$x_{k+1} = F(x_k, u_k)$	Nominal ego-traffic dynamics
$\tilde{x}_{k+1} = \tilde{F}(\tilde{x}_k, u_k, w_k)$	Stochastic dynamics model of the ego-traffic
$q, \phi, L(\tilde{x}) = \Sigma_{k=0}^{K-1} l(\tilde{x}_k)$	State cost, terminal cost, risk cost
α	The confidence level, tunes risk aversion
C_u	Admissible tail risk, hard cap on avg. tail severity
λ	MPPI temperature

We seek:

$$\min_v E\left[\phi(x_k) + \Sigma_{k=0}^{K-1}\left(q(x_k) + \frac{\lambda}{2}\ v_k^T \Sigma_\epsilon^{-1} v_k\right)\right]$$
$$s.t.\ CVaR_\alpha\big(L(\tilde{x})\big) \le C_u$$

Where, $CVaR_\alpha(L) = E[L \mid L \ge VaR_\alpha(L)]$ is the conditional expectation of the worst $1-\alpha$ tail

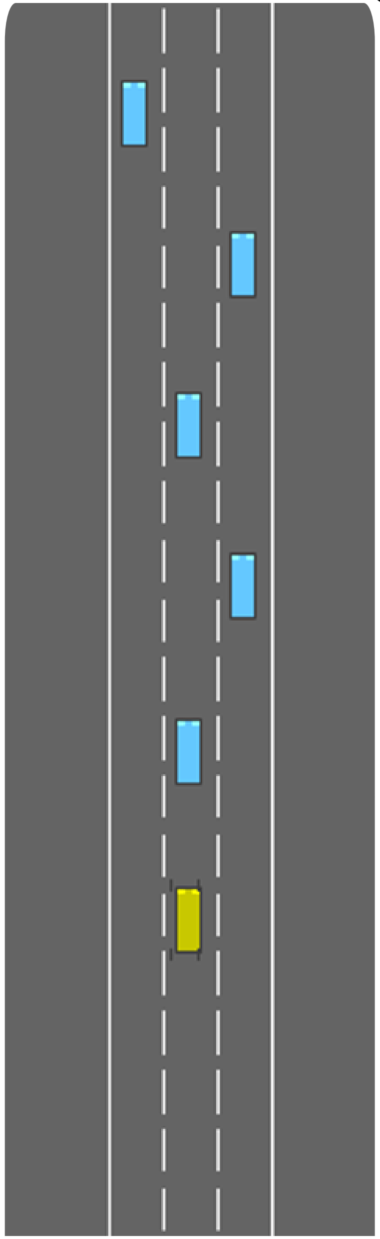
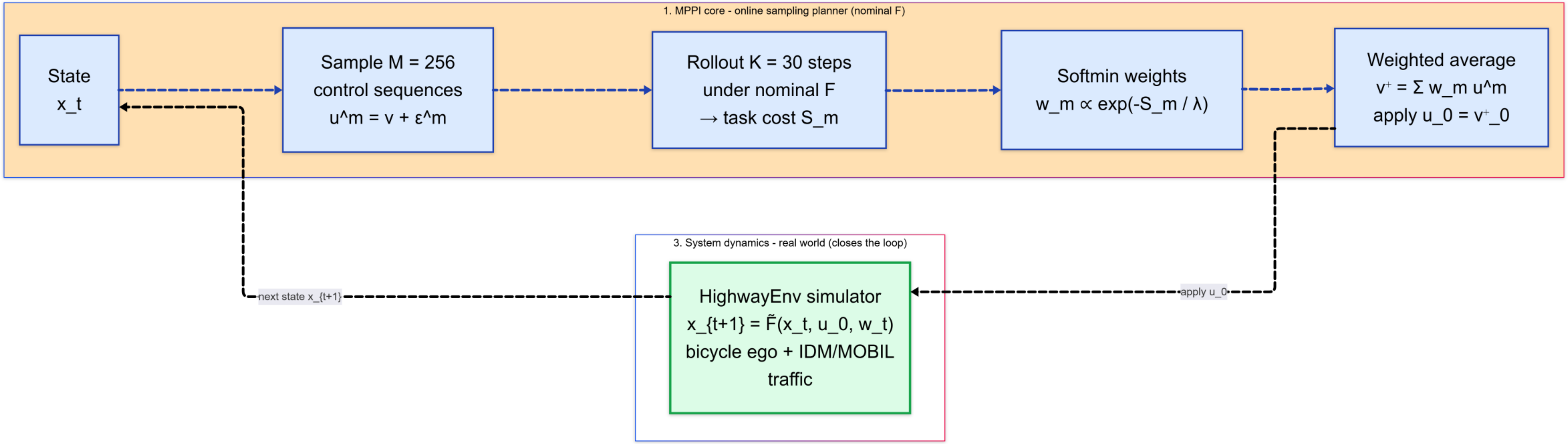


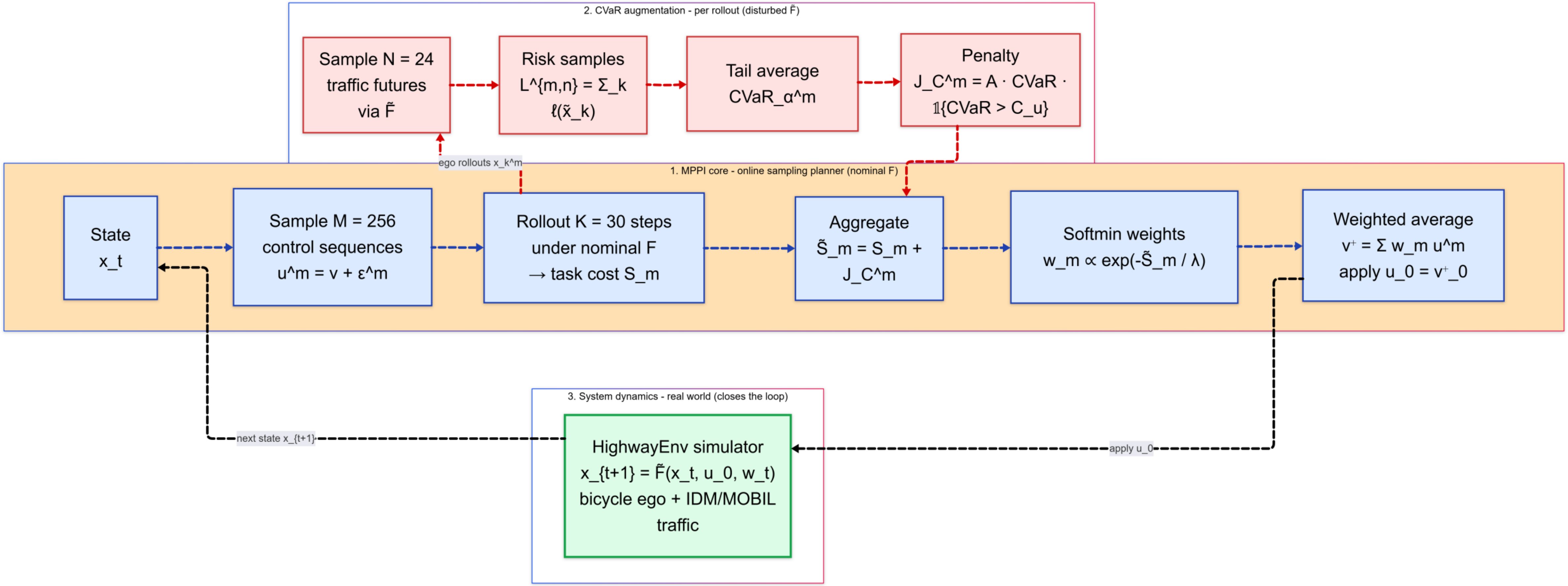
Fig. 2 HighwayEnv

Method: Model Predictive Path Integral (MPPI)



$$\min_{v} E\left[\phi(x_k) + \Sigma_{k=0}^{K-1}\left(q(x_k) + \frac{\lambda}{2}\ v_k^T \Sigma_{\epsilon}^{-1} v_k\right)\right]$$

Updated Method: Online CVaR integrated with MPPI

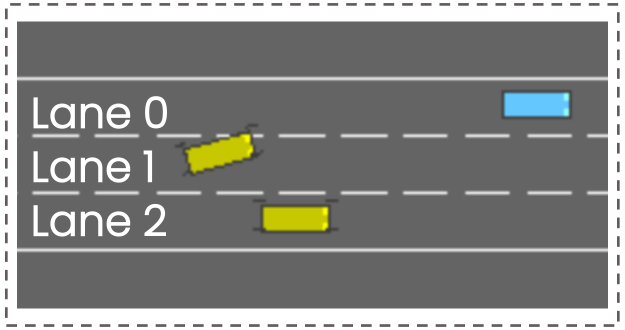


$$\min_{v} E\left[\phi(x_k) + \Sigma_{k=0}^{K-1}\left(q(x_k) + \frac{\lambda}{2}\ v_k^T \Sigma_\epsilon^{-1} v_k\right)\right]$$

$$s.t.\, CVaR_\alpha\left(L(\tilde{x})\right) \leq C_u$$

Simulation Environment, Scenarios, Performance Metrics (ISO)

Highway Layout



Performance Metrics

ISO	Metrics	Comfortable	Hard Limit
ISO-11270	Lateral acc $a_y, peak$ $[\frac{m}{s^2}]$	≤ 1.5	2.5
	Jerk $j_y, peak$ $[\frac{m}{s^3}]$	≤ 0.9	5.0
ISO-17387	Min. time to collision (TTC_{min}) (s)	≥ 4	2.0
	e_y, RMS $[m]$	≤ 0.05	0.25
	Time to lane change, T_{LC} (s)	$4 - 5$	Out of [2,8]

ISO-quality tuning deferred; project focus = CVaR mechanism validation

Highway Scenarios for Lane Change

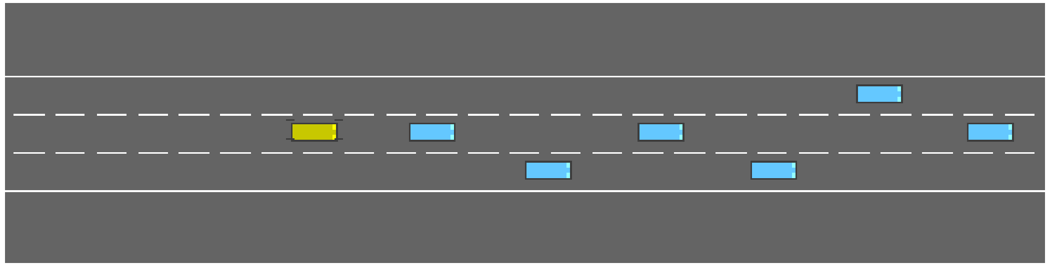


Fig 3. Lane Change: Lane 1 → Lane 0

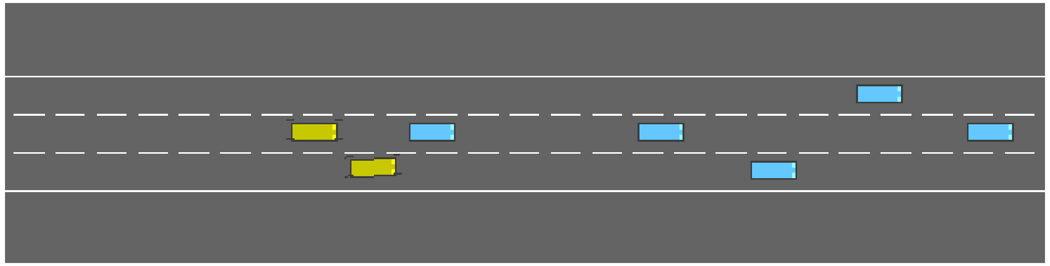


Fig 4. Cut-In: Lane 1 → Lane 1

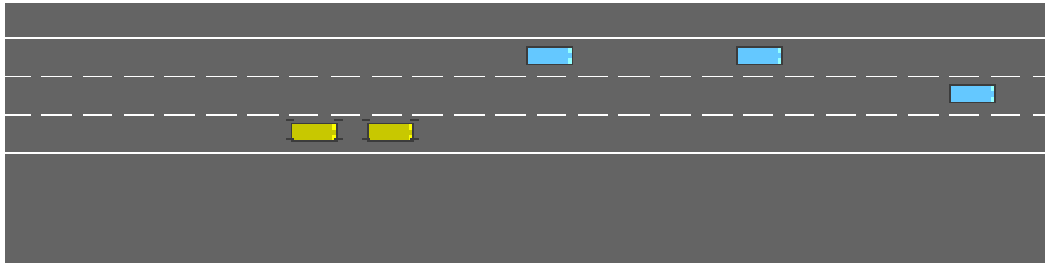


Fig 5. Cut-In: Lane 2 → Lane 0

MPPI VS CVaR-MPPI in Lane Change

Parameters	MPPI	CVaR-MPPI	Δ
Horizon (K)	30	30	
Samples (N)	256	256	
α	-	0.9	
MC Sample Size	-	64	
$\tilde{F}$ noise width	-	5	
Penalty Magnitude	-	200	
Performance Output			
$a_y, peak\ [\frac{m}{s^2}]$	15.21	17.07	+12.2%
$j_y, peak\ [\frac{m}{s^3}]$	95.34	68.36	-28.3%
TTC_{min}	0.01	0.75	+98.6%
e_y, RMS	0.762	0.219	-71.2%
T_{LC}	0.70	0.65	-0.05

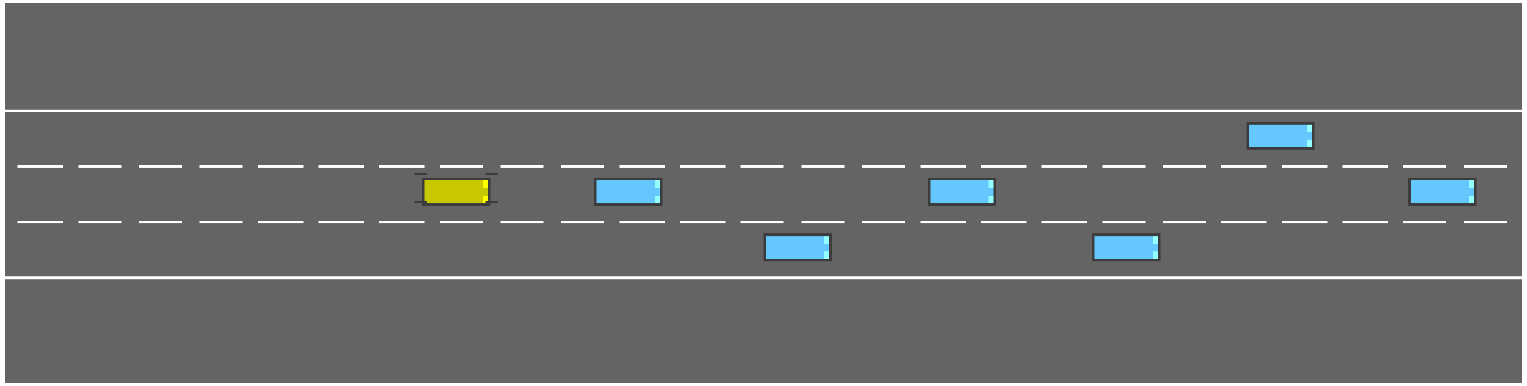


Fig 6. MPPI crashed in Lane Change scenario

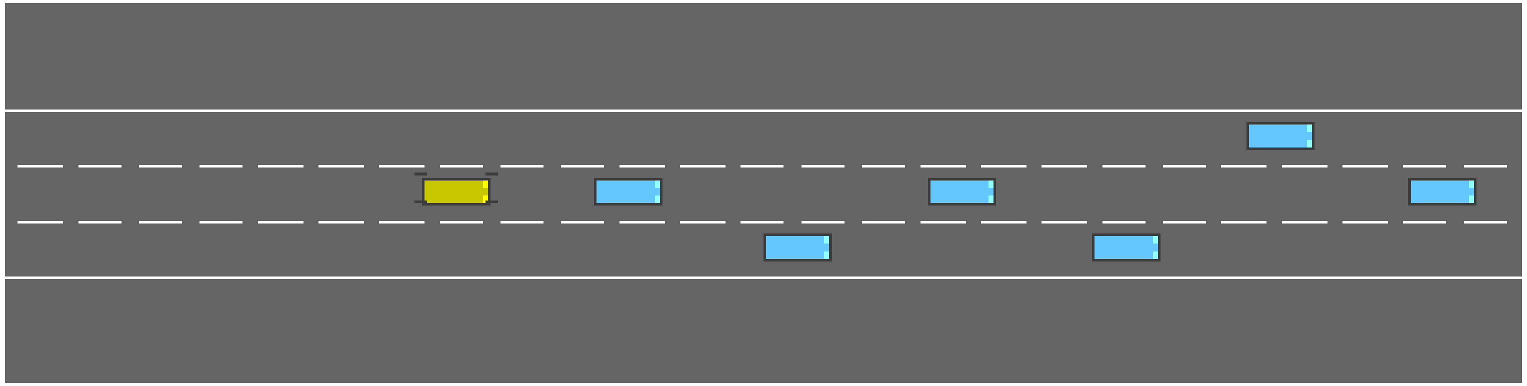
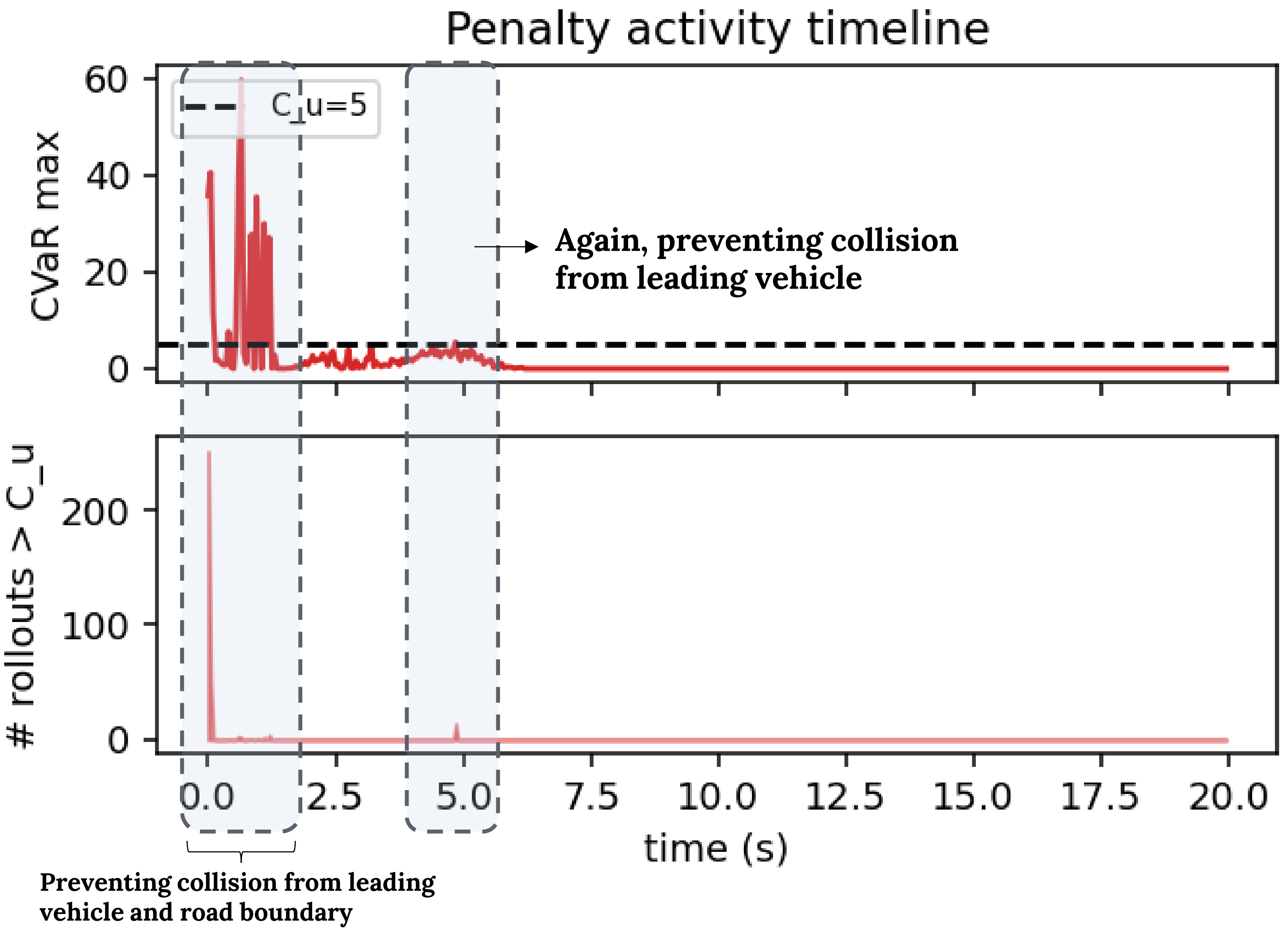


Fig 7. CVaR-MPPI prevented crash in Lane Change scenario

Evidence for CVaR in action

Runtime performance

- *227 ms / cycle on Intel i9-13900KF CPU (M=256, N=64, K=30). **"Online"** means evaluated at runtime*
- *Production-grade latency requires GPU vectorization, which is out of scope for this milestone.*



More Evidence: Analysis across the three scenarios

Scenario	Seeds	MPPI Crash	CVaR Crash	Notable per seed	Conclusion
Lane Change (Paired)	1	True	False	TTC: $0.01 \rightarrow 0.75$	Collision prevented
Hard Brake (paired)	1	False	False	$J_c = 0, \tilde{S} = 0$	No-regret proven
Cutin (sweep)	10	1/10	1/10	Seed 2: $\min\_dist$ 2.19 → 4.31	Margin gain
Lane Change (sweep)	10	1/10	0/10	Seed 2: MPPI collides where CVaR survived	Active risk aversion (More evidence required)

CVaR-MPPI delivers selective tail shaping — invisible when unneeded, decisive when needed, transparent when wrong

Lessons learned

CVaR helps only when $\widetilde{F}$ spans the hazard.

- Scripted out-of-distribution braking ⇒ silent; natural traffic perturbations ⇒ active.

No-regret is provable and verified.

$J_C \equiv 0 \Rightarrow \tilde{S} = S \; bit-for-bit$ (confirmed on hard braking scenario).

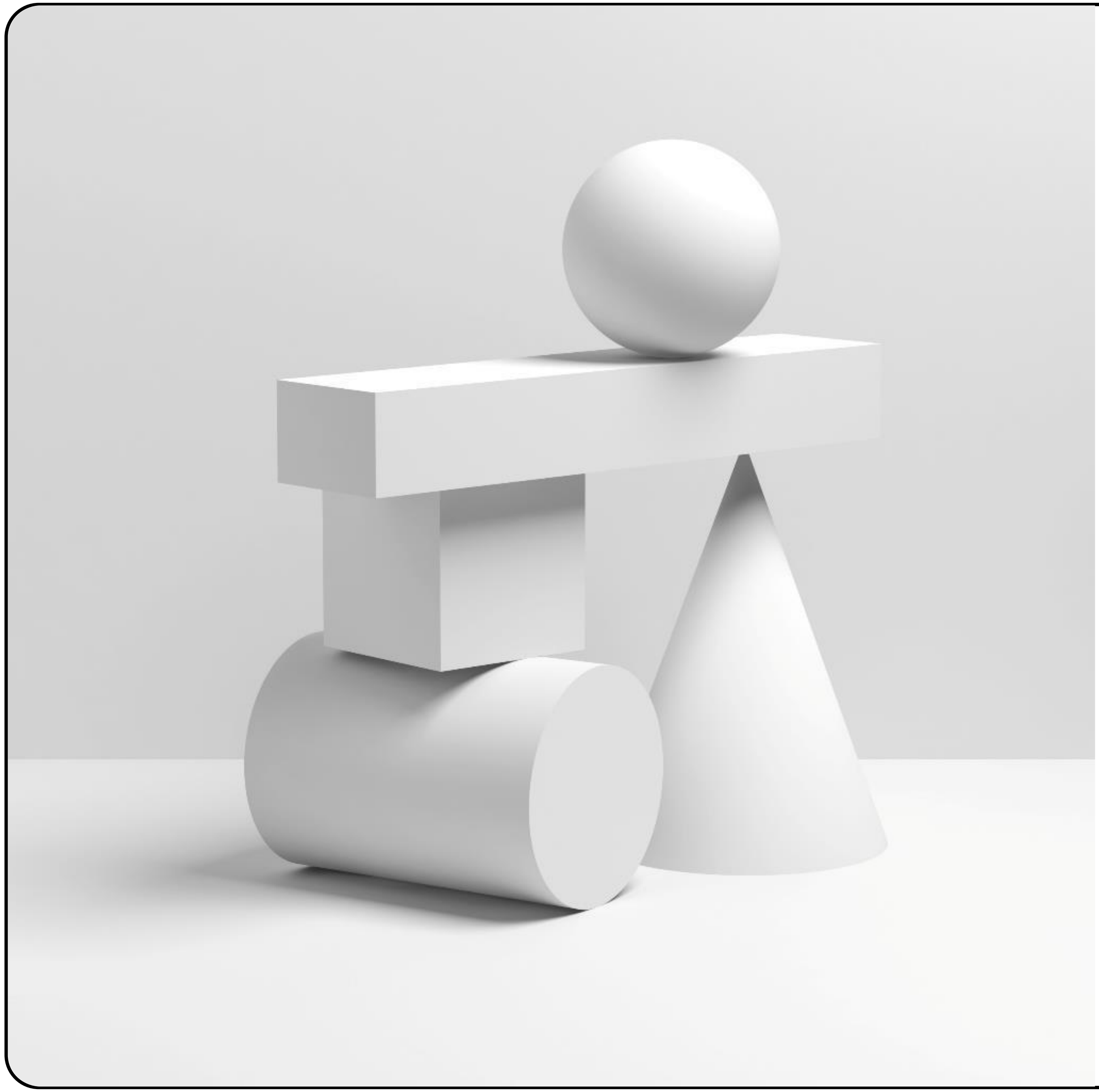
Calibrating C_u is the hardest knob.

Built per-step $\frac{n_{over-budget}}{CVaR_{\max}}$ logging so the penalty's activity is observable, not assumed.

Soft constraints can regress at the margin.

Future work: hard-constrained CVaR via Chow's confidence-set formulation.

Online CVaR-MPPI delivers selective tail shaping: invisible when unneeded, decisive when needed, transparent when wrong.



THANK YOU